

1. Small Web Application Specifications

Based on the AWS estimate and common small-scale application standards, the specifications are as follows:

Compute (CPU/RAM): 1 vCPU and 0.5 GB RAM (based on the t2.nano instance type seen in the image).

Storage Capacity: 10 GB (standard for small web application boot volumes and small S3/Blob storage needs).

Estimated Monthly Data Transfer: ~100 GB (standard baseline for small web traffic).

2. AWS Monthly Cost Estimate

Using result gotten from AWS:

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Compute: \$5.44 (using a t2.nano instance with a 3-year "No Upfront" Savings Plan in the London region).

Storage (S3): Standard S3 storage is approximately \$0.023 per GB, totaling \$0.23 for 10 GB.

Total AWS Estimate: ~\$5.67 per month.

3. Azure Monthly Cost Estimate

Comparing scenarios based on the provided Azure links:

Linux Scenario: Approximately \$4.00 – \$6.00 depending on the region and similar B-series (e.g., B1ls) instance sizing.

Windows Scenario (with Hybrid Benefit): Windows instances typically carry a premium, but with **Azure Hybrid Benefit**, you can save up to 40% on the license. Without the benefit, a similar Windows instance would cost ~\$12.00; with the benefit, it drops closer to the Linux price of ~\$7.50.

4. Networking: Inter-Zone Data Transfer

Networking costs can often be the "hidden" expense in cloud architecture.

AWS: Data transferred between Availability Zones (AZs) in the same region is generally charged at **\$0.01 per GB** in each direction.

Azure: Data transfer within the same Availability Zone is free, but inter-zone transfer (Availability Zone to Availability Zone) is also charged at roughly **\$0.01 per GB**.

Comparison: Both providers are neck-and-neck for inter-zone fees, but Azure offers more generous "Internet Egress" (outbound to the web) free tiers (100GB/month) compared to AWS's standard 100GB/month.

5. Discount Mechanisms

Both platforms offer significant discounts for long-term commitment:

AWS Savings Plans: Flexible pricing models that offer up to 72% savings compared to On-Demand prices in exchange for a commitment to a consistent amount of compute usage (e.g., \$1/hour) for a 1 or 3-year term.

Azure Reserved Instances (RIs): Allows you to reserve virtual machines for 1 or 3 years. This is more rigid than AWS Savings Plans (bound to specific instance families) but offers similar savings of up to 72%. Azure also offers a "Savings Plan for Compute" which mimics AWS's flexibility.

6. Comparison Report & Optimization

<i>Feature</i>	<i>AWS (London Region)</i>	<i>Azure (General)</i>	<i>Winner</i>
<i>Compute Price</i>	<i>\$5.44 (t2.nano)</i>	<i>~\$4.80 (B1ls)</i>	<i>Azure</i>
<i>Windows License</i>	<i>Full price included</i>	<i>Discounted via Hybrid Benefit</i>	<i>Azure</i>
<i>Flexibility</i>	<i>High (Savings Plans)</i>	<i>Moderate (RIs)</i>	<i>AWS</i>

Which provider is more cost-effective?

For Windows Workloads: Azure is significantly cheaper if you already own Windows Server licenses (via Hybrid Benefit).

For Ultra-Small Linux Apps: AWS provides very low-cost entry points like the t2.nano seen in your estimate.

Cost-Optimization Strategies:

Use Spot/Low-Priority Instances: For non-critical components of your web app (like background processing), use AWS Spot Instances or Azure Spot VMs to save up to 90% off On-Demand rates.

Right-Sizing: Monitor your CPU and RAM usage. If your t2.nano is constantly under 10% utilization, consider "serverless" options like AWS Lambda or Azure Functions, where you only pay for the milliseconds the code runs.